

Smoke and Carbon Monoxide Alarms

Where they go, how they are powered, and a distance sequence that makes sense once you know what causes false alarms.

Smoke alarm locations	Three of them: in each sleeping room, outside each sleeping area, and on each additional story including basements and habitable attics.
Interconnection	When one sounds, all sound. A fire in the basement has to wake someone asleep two floors up.
Power	Hard-wired to the building supply with battery backup.

The numbers here

Smoke alarm from the door of a bathroom containing a tub or shower	3 ft
Ionization alarm from a cooking appliance	20 ft
Ionization alarm fitted with a silencing switch	10 ft
Photoelectric alarm from a cooking appliance	6 ft

The sequence 20, 10, 6 is ordered by how prone each arrangement is to nuisance alarms. An ionization sensor reacts to the invisible particles that cooking produces in quantity, so it must stay furthest away. Give it a silencing button and it may come to ten feet, because the occupant can now stop it without taking the battery out. A photoelectric sensor barely reacts to cooking at all, so six feet is enough.

The reasoning matters more than the numbers: alarms that keep going off get disabled, and a disabled alarm protects nobody. Every one of these distances exists to stop that happening. The bathroom rule is the same idea — steam reads like smoke.

Carbon monoxide

When required	Where the dwelling has a fuel-fired appliance, or an attached garage with an opening communicating with the dwelling. Either condition on its own triggers it.
Where placed	Outside each separate sleeping area, in the immediate vicinity of the bedrooms.
The all-electric case	An all-electric house with no attached garage requires none — there is no combustion anywhere and nothing to produce carbon monoxide.

Say it back

Cover the page. Answer each one out loud.

1. Name all three required smoke alarm locations.
2. What does interconnection mean, and how must alarms be powered?
3. Give the three cooking-appliance distances and explain the order.
4. How far from a bathroom door, and why?
5. Under which two conditions is a CO alarm required?
6. How many CO alarms does an all-electric house with no garage need?